Calculus II

Lecture 03

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Basic Integration Rules

Example 5 (1/2):

Evaluate $\int \frac{\cos \theta}{\sin^2\theta}\, d\theta$

BASIC INTEGRATION RULES

Example 5 (2/2):

$$\text{Evaluate} \int \frac{\cos\theta}{\sin^2\theta}\, d\theta$$

$$= \int \left(\frac{1}{\sin\theta}\right)\left(\frac{\cos\theta}{\sin\theta}\right) d\theta$$

$$= \int \csc\theta \cot\theta \, d\theta = -\csc\theta + C$$

BASIC INTEGRATION RULES

Indefinite Integrals of Exponential Functions (1/2):

$$\int e^x \, dx = e^x + C$$

$$\int e^{kx} \, dx = \frac{e^{kx}}{k} + C, \quad k \neq 0$$

BASIC INTEGRATION RULES

Indefinite Integrals of Exponential Functions (2/2):

For $a > 0, a \neq 1$:

$$\int a^x \, dx = \frac{a^x}{\ln a} + C$$

$$\int a^{kx} \, dx = \frac{a^{kx}}{k(\ln a)} + C, \quad k \neq 0$$

BASIC INTEGRATION RULES

Example 1 (1/3):

$$\int 9e^t \, dt =$$

$$\int 2^{-5x} \, dx =$$

BASIC INTEGRATION RULES

Example 1 (2/3):

$$\int 9e^t \, dt = 9\int e^t \, dt = 9e^t + C$$

$$\int 2^{-5x} \, dx =$$

Basic Integration Rules

Example 1 (3/3):

$$\int 9e^t\,dt = 9\int e^t\,dt = 9e^t + C$$

$$\int 2^{-5x}\,dx = \frac{2^{-5x}}{-5(\ln 2)} + C = -\frac{2^{-5x}}{5(\ln 2)} + C$$

Basic Integration Rules

Indefinite Integrals of x^{-1}:

$$\int x^{-1}\,dx = \int \frac{1}{x}\,dx = \ln|x| + C$$

Basic Integration Rules

Examples:

$$\int \frac{4}{x}\,dx = 4\int \frac{1}{x}\,dx = 4\ln|x| + C$$

$$\int \left(-\frac{5}{x} + e^{-2x}\right)dx = -5\ln|x| - \frac{1}{2}e^{-2x} + C$$

Thank You

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